

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I Bhuvaneswar (192511402) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario Based Assignment

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1. Scenario

A hospital is designing an AI system to automatically generate monthly duty rosters for its nursing staff. The system must satisfy multiple constraints: no nurse should be scheduled for two overlapping shifts, a limited number of nurses are qualified for the ICU, certain senior nurses must supervise night shifts, nurses have individual leave and availability constraints, and special accommodations are needed for part-time staff. The complexity increases as the number of nurses and wards grows.

Tasks:

Formulate the problem as a Constraint Satisfaction Problem (CSP).

- i. Clearly define variables, domains, and constraints with explanation.
- ii. Explain how backtracking search works in this scenario.
- iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.
- iv. Analyze real-world challenges and justify the best strategy for scalability.

2. Scenario

A mobile gaming company is developing an AI opponent for a chess app aimed at casual players. The AI must compete against human players, make decisions within strict time limits (a few seconds per move), evaluate complex board positions with many possible moves, and the decision tree is extremely large, making computation expensive.

Tasks:

- i. Explain how the Minimax algorithm models this problem.
- ii. Analyze the computational challenges of large game trees.
- iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.
- iv. Design an evaluation function and justify the factors considered.
- v. Recommend a strategy for balancing optimality and real-time performance.

3. Scenario

A disaster-response agency deploys AI-guided drones to airlift medical supplies to villages cut off by a wildfire. The environment is highly dynamic - access roads and forest trails may become impassable without warning, and smoke conditions affect travel cost. The drone has access to: a heuristic estimate (straight-line distance to each village), partial real-time updates about newly blocked trails, and variable movement costs depending on terrain and visibility. Due to the urgency of the situation, the drone must quickly decide routes with minimum delay and risk.

Tasks:

- Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.
- Explain how A* would handle the same situation, including how it balances cost and heuristic.
- Analyze the impact of heuristic accuracy on both algorithms.
- Discuss how dynamic changes (newly blocked trails) affect search performance.
- Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

$$20 \times 20 + 22 \times 18 + 12$$

= 92

PF

Scenario:

hospital is designing an AI system:

Tasks:

Formulation as a CSP:

A Constraint Satisfaction Problem consists of variables, domains and constraints.

Constraints:

- * No overlapping shifts
- * Only qualified nurses can work in ICU
- * Senior nurses must supervise night shifts.
- * Nurses cannot be assigned during leave.

(ii) Backtracking Search:

Backtracking assigns shifts one by one. If an assignment violates a constraint, it undoes the assignment and tries another option. This continues until a valid roster is obtained.

(iii) Constraint Propagation:

Forward checking removes invalid choices from future assignments whenever a constraint is detected.

(i) Real-world challenges & Scalability:

large hospitals have many nurses, wards, shift leave request and skill requirement. Therefore backtracking with forward checking and heuristics such as MPR.

(2) Scenario

A mobile gaming company:

Tasks:

(i) Minimax Algorithm:

Minimax models chess as a two player games.

- * AI tries to maximize its score.
- * Human player tries to minimize the AI's score.
- * The algorithm explores possible moves and selects the best move.

(ii) Computational challenges:

Chess has very large game tree with many possible moves. Searching the entire tree requires high computation and time, which is difficult for real-time play.

Alpha - Beta Pruning

Alpha - Beta Pruning improves Minimax by removing branches that cannot affect the final decision.

Alpha - Best value found for the maximizing player.

Beta: Best value found for the minimizing player.
when $\alpha \geq \beta$, the remaining branches is pruned.

Evaluation Function:

The evaluation functions can consider:

Material : value of Pieces

King safety

Piece Positions

Mobility : Number of possible moves

Pawn structure.

(v) Best Strategy:

Use Minimax with alpha-Beta pruning and depth / time limit. This provides a good balance between playing strength, optimality, and real time performance.

② Scenario:

A disaster-response agency deploys AI guided drones

Tasks:

(i) Greedy Best-First Search:

Greedy Best-First Search selects the path that appears closest to the destination using the heuristic.

Advantage:

* Fast and Simple

* Useful when quick decision are required.

Search:

(i) * uses both the actual path cost and the heuristic:

$$f(n) = g(n) + h(n)$$

* $g(n)$ = cost from the starting point

* $h(n)$ = estimated cost to the destination.

(ii) Impact of Heuristic Accuracy:

* An accurate heuristic helps both algorithms find better routes quickly.

* A poor heuristic can make the search slower or lead to inefficient routes.

(iv) Dynamic Changes:

If a trail becomes blocked, the previously calculated route may become invalid. The drone must update the search and find a new available path.

Recommended Algorithm:

A* is the better choice because it considers both travel cost and heuristic distance. It provides a good balance between speed, route quality and safety, making it suitable for the dynamic real-time environment.
